INTERNATIONAL MONETARY FUND

MALAWI

April 27, 2020

REQUEST FOR DISBURSEMENT UNDER THE RAPID CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS[1]

Approved By
David Robinson and S. Ali Abbas (IMF) and Marcello Estevão (IDA)

Prepared by the staffs of International Monetary Fund and the International Development Association (IDA)

Malawi: Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	*Moderate*[1]
Overall risk of debt distress	*High*
Granularity in the risk rating	*Some space to absorb shocks*
Application of judgement	*No*
Macroeconomic projections	Growth is expected to fall to 1 percent in 2020 and to 2.5 percent in 2021—assuming it will take time for businesses to re-open after the partial lockdown ends and for trade flows to normalize. An economic recovery is expected thereafter, with growth averaging 6.4 percent during 2022-25. Over the long-term, growth is expected to stabilize around 5.5 percent.
Financing strategy	External financing in the form of concessional project loans is expected to gradually increase. There is a risk that the pace of this increase may accelerate if the economy's absorption capacity increases faster than expected. The grant element of project loans will remain relatively high over the forecast period, with no access to market financing.

[1] The analysis presented in this document is based on the debt sustainability framework for low-income countries approved by the Boards of both the International Monetary Fund and the International Development Association. It has been prepared in the context of the 2020 request for emergency financing under the IMF's Rapid Credit Facility. The macroeconomic framework underlying this DSA update is the same as that included in the staff report of the 2020 RCF request which reflects currently available information, including recent global and domestic developments. However, updates with respect to the economic impact and policy response to the COVID-19 pandemic are rapidly evolving and risks are heavily tilted to the downside.

Malawi: Joint Bank-Fund Debt Sustainability Analysis (Concluded)	
Mechanical risk rating under the external DSA	Moderate
Mechanical risk rating under the public DSA	High
[1]Malawi's debt carrying capacity is classified as "medium" according to the composite indicator (CI) score. Malawi's CI score based on the current vintage (2019 CPIA and 2019 October WEO) is 2.84 and was 2.72 in the previous vintage (2018 CPIA and 2019 April WEO)—both above the threshold value of 2.69—resulting in an upgrade of the debt carrying capacity as the CI score has been above the threshold for two consecutive vintages.	

Malawi is at moderate risk of external debt distress with some space to absorb shocks and high risk of overall debt distress. Under the baseline scenario reflecting the macroeconomic impact and rise in external borrowing in 2020 related to the COVID-19 pandemic and a gradual increase in project loans over the medium term, a moderate rating is maintained for external debt distress with debt carrying capacity upgraded to "medium"—revision in October 2019. The present value of total public debt to GDP is projected to exceed the benchmark for the near and medium terms and then gradually decline under the baseline scenario. This mainly reflects larger primary deficits during FY 2019/20-20/21 resulting in increasing amounts of domestic debt.

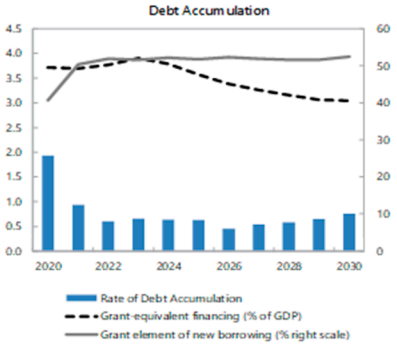
This DSA incorporates current projections of the impact from the COVID-19 pandemic: (i) GDP growth is expected to fall to 1 percent in 2020 and to 2.5 percent in 2021 as a lockdown substantially slows domestic activity, especially in domestic manufacturing and wholesale and retail trade, and the global and resultant regional economic slowdown is weighing on exports, remittances, and foreign direct investment (FDI); (ii) the domestic primary balance is expected to deteriorate (relative to pre-pandemic projections) to -3.4 percent of GDP in FY 2019/20 and -1.3 percent of GDP in FY 2020/21; (iii) the current account deficit is expected to widen to 18.1 percent of GDP in 2020 and remain broadly unchanged in 2021. However, the economic impact of the COVID-19 pandemic and policy responses are rapidly evolving and risks are heavily tilted to the downside, which could lead to a faster-than-expected deterioration in external and public debt indicators.

Table 1. Malawi: External Debt Sustainability Framework, Baseline Scenario, 2017–2040

(in percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/ 9/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
External debt (nominal) 1/	36.5	34.6	32.8	34.4	36.1	35.8	35.6	35.3	35.1	30.8	30.2	30.5	33.9
of which: public and publicly guaranteed (PPG)	32.8	31.2	29.6	31.4	33.0	32.8	32.6	32.4	32.2	28.4	28.6	26.9	31.1
Change in external debt	0.7	-1.9	-1.9	1.6	1.7	-0.2	-0.3	-0.2	-0.2	-0.4	0.1		
Identified net debt-creating flows	16.5	15.4	12.4	16.3	15.7	13.1	12.9	12.2	11.3	10.3	7.7	11.4	12.3
Non-interest current account deficit	22.0	20.2	16.9	17.5	17.6	16.7	16.4	15.8	15.1	14.1	11.5	13.6	15.6
Deficit in balance of goods and services	27.3	26.3	23.1	22.6	23.0	22.3	22.1	21.4	20.8	19.7	17.3	22.4	21.2
Exports	16.9	16.1	16.3	13.8	15.0	15.7	16.4	17.0	17.7	18.7	21.0		
Imports	44.2	42.4	39.4	36.5	38.0	38.0	38.4	38.5	38.5	38.3	38.3		
Net current transfers (negative = inflow)	-7.5	-8.1	-8.3	-7.0	-7.3	-7.6	-7.7	-7.8	-7.9	-7.7	-7.5	-6.5	-7.7
of which: official	-0.2	0.0	-0.5	-0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other current account flows (negative = net inflow)	2.2	2.1	2.1	1.9	2.0	2.0	2.1	2.1	2.2	2.1	1.8	-2.2	2.1
Net FDI (negative = inflow)	-1.5	-1.6	-1.4	-1.2	-1.5	-1.6	-1.6	-1.7	-1.9	-2.4	-2.5	-1.8	-1.9
Endogenous debt dynamics 2/	-4.0	-3.3	-3.1	0.0	-0.5	-2.0	-1.9	-1.9	-1.8	-1.3	-1.3		
Contribution from nominal interest rate	0.3	0.3	0.3	0.3	0.4	0.3	0.3	0.3	0.3	0.3	0.3		
Contribution from real GDP growth	-1.3	-1.0	-1.4	-0.3	-0.9	-2.3	-2.2	-2.1	-2.1	-1.6	-1.5		
Contribution from price and exchange rate changes	-3.0	-2.6	-2.0	...	...	...	...	...	...	...	...		
Residual 3/	-15.8	-17.3	-14.3	-14.6	-14.0	-13.3	-13.1	-12.4	-11.6	-10.7	-7.6	-9.7	-12.5
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	...	...	18.8	18.9	19.8	19.8	19.7	19.5	19.3	16.5	16.8		
PV of PPG external debt-to-exports ratio	...	...	115.4	136.3	132.1	126.0	120.4	114.4	109.1	88.2	79.8		
PPG debt service-to-exports ratio	9.3	12.1	6.5	8.3	8.8	9.5	8.9	8.6	8.3	6.5	4.4		
PPG debt service-to-revenue ratio	8.0	9.7	5.6	6.6	6.9	7.6	7.3	7.3	7.3	6.0	6.0		
Gross external financing need (Billion of U.S. dollars)	1.4	1.4	1.3	1.5	1.5	1.5	1.5	1.5	1.5	1.8	2.6		
Key macroeconomic assumptions													
Real GDP growth (in percent)	4.0	3.2	4.5	1.0	2.5	6.5	6.5	6.3	6.3	5.5	5.5	4.1	5.2
GDP deflator in US dollar terms (change in percent)	9.1	7.5	6.1	8.9	-2.3	-3.2	-2.7	-1.8	-1.8	1.1	1.7	-1.1	0.2
Effective interest rate (percent) 4/	0.9	0.9	0.9	0.9	1.2	0.8	0.9	0.8	0.8	0.9	1.0	0.8	0.9
Growth of exports of G&S (US dollar terms, in percent)	-10.7	5.6	12.4	-6.7	8.3	8.0	8.1	8.6	8.3	7.7	8.7	5.8	6.7
Growth of imports of G&S (US dollar terms, in percent)	13.0	6.2	3.2	1.8	4.2	3.2	4.8	4.5	4.5	6.5	7.5	4.7	5.1
Grant element of new public sector borrowing (in percent)	...	...	...	40.6	50.3	51.9	51.6	52.2	51.8	52.5	48.8	...	50.8
Government revenues (excluding grants, in percent of GDP)	19.5	20.2	18.8	17.4	19.2	19.6	20.0	20.0	20.0	20.5	15.5	18.4	19.9
Aid flows (in Billion of US dollars) 5/	0.4	0.2	0.4	0.3	0.4	0.4	0.4	0.4	0.4	0.5	0.7		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	3.7	3.7	3.8	3.9	3.8	3.6	3.0	2.0	...	3.5
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	64.5	73.8	77.1	77.5	77.1	75.9	72.7	59.5	...	74.1
Nominal GDP (Billion of US dollars)	6.2	6.9	7.7	8.4	8.4	8.7	9.0	9.4	9.8	13.5	26.3		
Nominal dollar GDP growth	13.4	10.9	10.9	10.0	0.1	3.1	3.7	4.4	4.4	6.7	7.3	3.1	5.3
Memorandum items:													
PV of external debt 7/	...	...	22.0	21.8	22.8	22.8	22.7	22.4	22.2	18.9	18.4		
In percent of exports	...	...	134.9	157.8	152.5	145.4	138.8	131.8	125.5	101.2	87.6		
Total external debt service-to-exports ratio	9.3	12.1	8.6	10.5	10.9	11.4	10.7	10.2	9.8	7.6	4.9		
PV of PPG external debt (in Billion of US dollars)			1.4	1.6	1.7	1.7	1.8	1.8	1.9	2.2	4.4		
(PVt-PVt-1)/GDPt-1 (in percent)				1.9	0.9	0.6	0.7	0.6	0.6	0.8	1.3		
Non-interest current account deficit that stabilizes debt ratio	21.3	22.1	18.8	15.9	16.0	17.0	16.7	16.0	15.3	14.5	11.4		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	Yes



Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

9/ The suspension of debt service to Malawi's official bilateral creditors under the G20 Initiative is not incorporated due to a lack of sufficient information at this time.

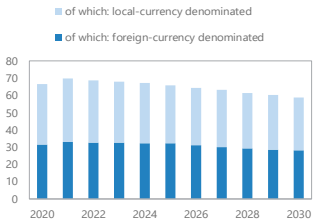
Table 2. Malawi: Public Sector Debt Sustainability Framework, Baseline Scenario, 2017–2040

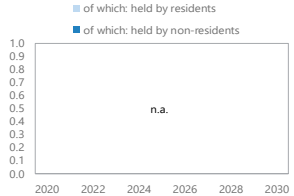
(in percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2017	2018	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
Public sector debt 1/	**56.7**	**59.7**	**59.5**	**66.5**	**69.9**	**68.8**	**68.0**	**67.4**	**65.8**	**58.7**	**45.2**	**47.2**	**64.9**
of which: external debt	**32.8**	**31.2**	**29.6**	**31.4**	**33.0**	**32.8**	**32.6**	**32.4**	**32.2**	**28.4**	**28.6**	**26.9**	**31.1**
Change in public sector debt	1.6	3.0	-0.1	7.0	3.3	-1.0	-0.8	-0.6	-1.6	-1.4	-0.7		
Identified debt-creating flows	**0.3**	**0.4**	**-0.1**	**5.3**	**3.4**	**-0.6**	**-0.5**	**-0.3**	**-1.2**	**-1.2**	**-0.7**	**0.6**	**0.1**
Primary deficit	**3.5**	**2.1**	**2.2**	**4.8**	**3.7**	**2.5**	**2.0**	**2.2**	**1.6**	**-0.3**	**0.6**	**1.5**	**1.6**
Revenue and grants	22.7	21.5	21.2	19.7	21.6	22.2	22.7	22.6	22.4	22.3	16.3	**22.6**	**22.1**
of which: grants	3.3	1.3	2.3	2.3	2.4	2.6	2.7	2.6	2.4	1.8	0.7		
Primary (noninterest) expenditure	26.3	23.6	23.3	24.5	25.3	24.8	24.8	24.8	24.0	22.0	16.9	**24.1**	**23.7**
Automatic debt dynamics	**-3.2**	**-1.7**	**-2.3**	**0.5**	**-0.3**	**-3.2**	**-2.5**	**-2.5**	**-2.8**	**-0.9**	**-1.3**		
Contribution from interest rate/growth differential	-0.8	-0.2	-1.0	0.4	-1.5	-4.4	-3.7	-3.7	-3.9	-1.1	-1.4		
of which: contribution from average real interest rate	1.3	1.6	1.6	1.0	0.1	-0.2	0.5	0.4	0.0	2.0	1.0		
of which: contribution from real GDP growth	-2.1	-1.7	-2.6	-0.6	-1.6	-4.3	-4.2	-4.0	-4.0	-3.1	-2.4		
Contribution from real exchange rate depreciation	-2.4	-1.6	-1.3	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**1.3**	**2.6**	**0.0**	**1.7**	**1.1**	**0.8**	**0.8**	**0.8**	**0.8**	**0.0**	**0.1**	**2.0**	**0.5**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**48.7**	**55.1**	**57.7**	**56.6**	**55.7**	**55.1**	**53.4**	**47.0**	**33.7**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**229.9**	**280.2**	**267.4**	**255.0**	**245.4**	**243.7**	**238.7**	**211.2**	**207.1**		
Debt service-to-revenue and grants ratio 3/	**35.6**	**25.3**	**27.8**	**34.9**	**53.3**	**64.7**	**71.1**	**76.4**	**80.3**	**97.8**	**84.6**		
Gross financing need 4/	7.6	3.6	8.0	11.7	15.2	16.9	18.2	19.5	19.6	21.4	14.4		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	4.0	3.2	4.5	1.0	2.5	6.5	6.5	6.3	6.3	5.5	5.5	**4.1**	**5.2**
Average nominal interest rate on external debt (in percent)	1.0	1.1	1.0	1.0	1.4	1.0	1.1	0.9	0.9	1.0	1.1	**1.0**	**1.0**
Average real interest rate on domestic debt (in percent)	6.6	8.4	7.2	4.6	1.0	0.5	2.4	2.1	1.1	7.6	7.3	**3.6**	**4.3**
Real exchange rate depreciation (in percent, + indicates depreciation)	-8.0	-4.9	-4.3	...	...	...	...	...	...	...	...	**5.4**	**...**
Inflation rate (GDP deflator, in percent)	10.9	7.8	8.0	13.2	9.5	6.4	4.5	4.4	4.4	4.4	4.9	**16.0**	**5.9**
Growth of real primary spending (deflated by GDP deflator, in percent)	16.2	-7.2	3.3	6.0	5.9	4.2	6.6	6.5	2.7	2.8	0.3	**3.5**	**4.6**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	1.9	-0.9	2.3	-2.2	0.4	3.6	2.9	2.9	3.2	1.1	1.3	**1.1**	**1.7**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**Yes**

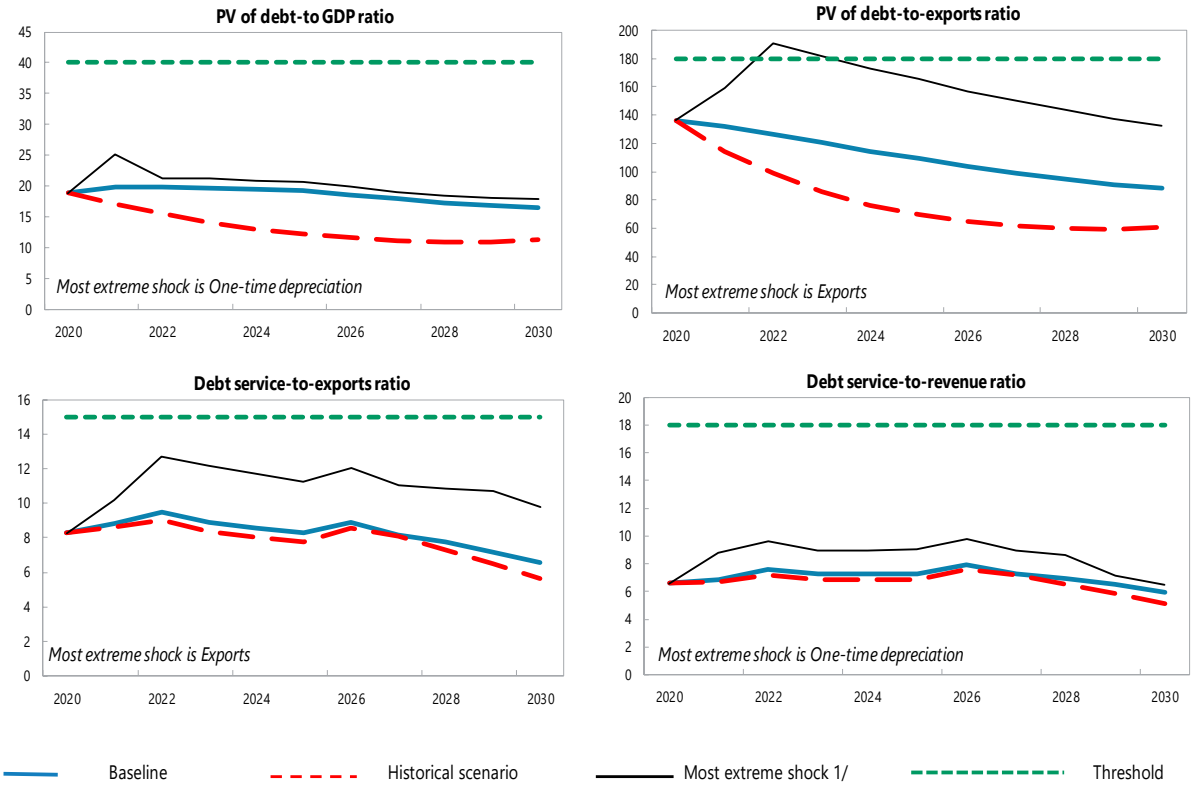
Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.
1/ Coverage of debt: The central government, central bank, government-guaranteed debt. Definition of external debt is Currency-based.
2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.
4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.
5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.
6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Malawi: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2020–2030



Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	Yes	
Natural Disasters	No	No
Commodity Prices 2/	n.a.	n.a.
Market Financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	0.8%	0.8%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	35	35
Avg. grace period	6	6

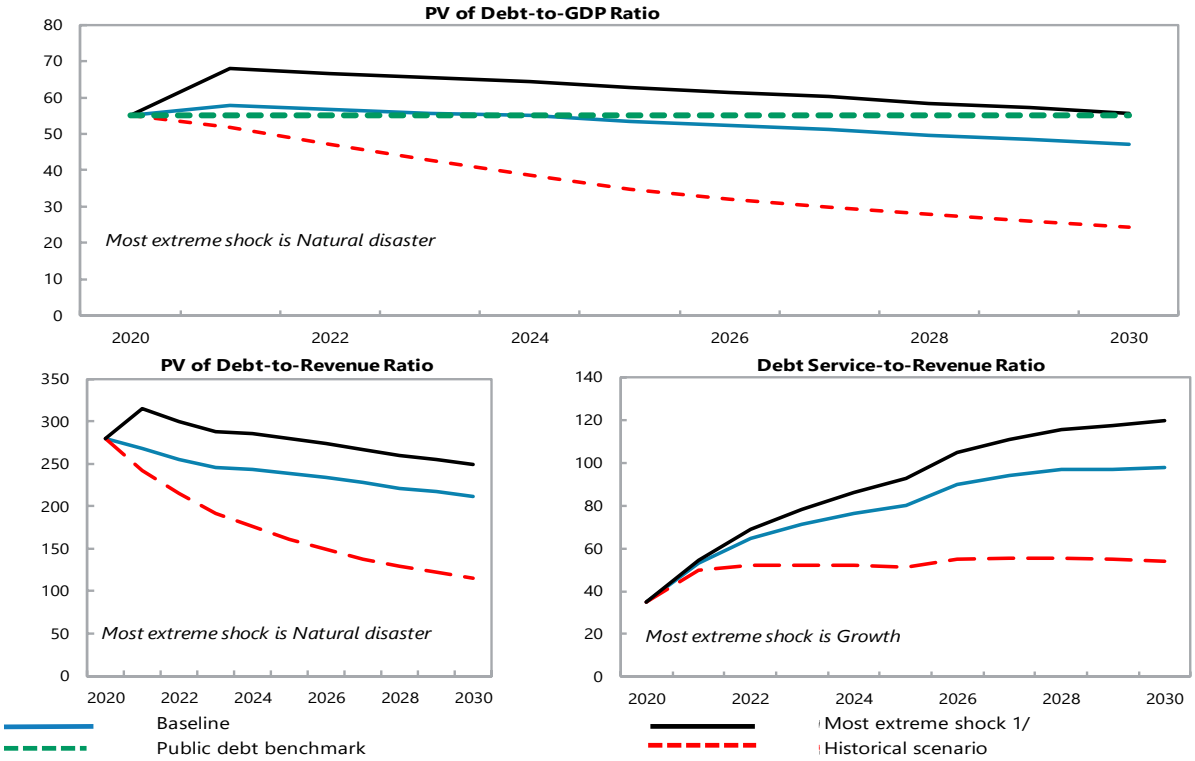
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Malawi: Indicators of Public Debt Under Alternative Scenarios, 2020–2030



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	14%	14%
Domestic medium and long-term	26%	26%
Domestic short-term	60%	60%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	0.8%	0.8%
Avg. maturity (incl. grace period)	35	35
Avg. grace period	6	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	7.0%	7.0%
Avg. maturity (incl. grace period)	2	2
Avg. grace period	1	1
Domestic short-term debt		
Avg. real interest rate	3%	3.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Malawi: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2020–2030

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of debt-to GDP ratio											
Baseline	18.9	19.8	19.8	19.7	19.5	19.3	18.6	17.9	17.3	16.8	16.5
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	18.9	17.1	15.5	14.1	13.0	12.3	11.6	11.2	10.9	11.0	11.4
B. Bound Tests											
B1. Real GDP growth	18.9	20.3	20.8	20.7	20.5	20.3	19.6	18.9	18.2	17.7	17.3
B2. Primary balance	18.9	19.9	20.1	20.2	20.1	20.0	19.5	18.8	18.3	17.8	17.5
B3. Exports	18.9	20.9	22.9	22.8	22.6	22.4	21.7	20.9	20.2	19.5	18.9
B4. Other flows 3/	18.9	21.5	23.5	23.4	23.2	23.0	22.3	21.5	20.7	19.9	19.4
B5. One-time 30 percent nominal depreciation	18.9	25.2	21.3	21.2	20.9	20.6	19.9	19.1	18.4	18.1	17.9
B6. Combination of B1-B5	18.9	22.1	23.1	23.0	22.8	22.5	21.9	21.1	20.3	19.6	19.1
C. Tailored Tests											
C1. Combined contingent liabilities	18.9	20.4	20.8	21.1	21.1	21.2	20.8	20.2	19.7	19.3	19.1
C2. Natural disaster	18.9	20.8	21.2	21.6	21.7	21.9	21.5	21.1	20.7	20.4	20.2
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	136.3	132.1	126.0	120.4	114.4	109.1	103.4	98.6	94.5	91.0	88.2
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	136.3	114.3	98.8	86.0	76.2	69.4	64.5	61.4	59.7	59.3	61.1
B. Bound Tests											
B1. Real GDP growth	136.3	132.1	126.0	120.4	114.4	109.1	103.4	98.6	94.5	91.0	88.2
B2. Primary balance	136.3	132.9	128.1	123.3	118.0	113.2	108.0	103.6	99.7	96.4	93.7
B3. Exports	136.3	159.4	**190.4**	**182.1**	173.2	165.3	157.1	150.1	143.7	137.4	132.4
B4. Other flows 3/	136.3	143.9	149.6	143.1	136.2	130.0	123.6	118.2	112.9	107.9	103.7
B5. One-time 30 percent nominal depreciation	136.3	132.1	106.5	101.6	96.4	91.8	86.7	82.5	78.9	76.6	75.1
B6. Combination of B1-B5	136.3	155.7	141.6	158.6	150.9	144.0	136.7	130.6	124.8	119.5	115.2
C. Tailored Tests											
C1. Combined contingent liabilities	136.3	136.2	132.3	128.6	124.1	119.9	115.2	111.2	107.7	104.6	102.2
C2. Natural disaster	136.3	141.3	137.6	134.3	130.1	126.2	121.8	118.1	114.9	112.2	110.2
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	8.3	8.8	9.5	8.9	8.6	8.3	8.9	8.2	7.7	7.2	6.5
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	8.3	8.6	9.0	8.4	8.0	7.8	8.6	8.1	7.3	6.5	5.7
B. Bound Tests											
B1. Real GDP growth	8.3	8.8	9.5	8.9	8.6	8.3	8.9	8.2	7.7	7.2	6.5
B2. Primary balance	8.3	8.8	9.5	8.9	8.6	8.3	9.0	8.2	7.9	7.3	6.7
B3. Exports	8.3	10.2	12.7	12.1	11.7	11.3	12.0	11.1	10.8	10.7	9.8
B4. Other flows 3/	8.3	8.8	9.7	9.3	8.9	8.6	9.2	8.4	8.5	8.4	7.6
B5. One-time 30 percent nominal depreciation	8.3	8.8	9.5	8.6	8.3	8.0	8.6	7.9	7.5	6.2	5.6
B6. Combination of B1-B5	8.3	9.6	11.3	10.7	10.3	9.9	10.6	9.8	9.8	9.3	8.5
C. Tailored Tests											
C1. Combined contingent liabilities	8.3	8.8	9.6	9.0	8.7	8.4	9.0	8.3	7.9	7.4	6.7
C2. Natural disaster	8.3	9.1	9.9	9.3	9.0	8.7	9.4	8.6	8.2	7.7	7.0
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	6.6	6.9	7.6	7.3	7.3	7.3	7.9	7.2	6.9	6.5	6.0
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	6.6	6.7	7.2	6.8	6.8	6.9	7.6	7.2	6.6	5.9	5.1
B. Bound Tests											
B1. Real GDP growth	6.6	7.1	8.0	7.7	7.7	7.7	8.3	7.6	7.3	6.8	6.3
B2. Primary balance	6.6	6.9	7.6	7.3	7.3	7.4	8.0	7.3	7.0	6.6	6.1
B3. Exports	6.6	6.9	7.8	7.6	7.6	7.6	8.2	7.5	7.4	7.4	6.8
B4. Other flows 3/	6.6	6.9	7.7	7.6	7.6	7.6	8.2	7.5	7.6	7.6	7.0
B5. One-time 30 percent nominal depreciation	6.6	8.8	9.7	9.0	9.0	9.0	9.8	9.0	8.6	7.1	6.5
B6. Combination of B1-B5	6.6	7.1	8.0	7.8	7.8	7.8	8.4	7.7	7.8	7.5	6.9
C. Tailored Tests											
C1. Combined contingent liabilities	6.6	6.9	7.6	7.4	7.4	7.4	8.1	7.4	7.1	6.7	6.1
C2. Natural disaster	6.6	6.9	7.6	7.4	7.4	7.5	8.1	7.4	7.1	6.7	6.2
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Malawi: Sensitivity Analysis for Key Indicators of Public Debt, 2020–2030

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of Debt-to-GDP Ratio											
Baseline	**55.1**	**57.7**	**56.6**	**55.7**	**55.1**	53.4	52.5	51.3	49.6	48.5	47.0
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	**55**	52	47	43	39	35	32	30	28	26	24
B. Bound Tests											
B1. Real GDP growth	**55**	**60**	**61**	**61**	**61**	**60**	**59**	**59**	**58**	**57**	**56**
B2. Primary balance	**55**	**59**	**60**	**59**	**58**	**56**	**55**	54	52	51	49
B3. Exports	**55**	**59**	**60**	**59**	**58**	**56**	**55**	54	52	51	49
B4. Other flows 3/	**55**	**60**	**60**	**60**	**59**	**57**	**56**	55	53	52	50
B5. One-time 30 percent nominal depreciation	**55**	**58**	**55**	54	52	50	48	46	43	42	39
B6. Combination of B1-B5	**55**	**57**	**56**	54	53	52	51	49	48	47	45
C. Tailored Tests											
C1. Combined contingent liabilities	**55**	**66**	**64**	**63**	**62**	**60**	**59**	**57**	**55**	54	52
C2. Natural disaster	**55**	**68**	**67**	**65**	**64**	**63**	**61**	**60**	**58**	**57**	**56**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	280.2	267.4	255.0	245.4	243.7	238.7	234.0	227.6	221.0	217.1	211.2
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	280	242	215	192	176	161	149	138	130	122	115
B. Bound Tests											
B1. Real GDP growth	280	275	272	265	267	266	264	260	256	255	252
B2. Primary balance	280	275	270	259	257	251	246	239	231	227	221
B3. Exports	280	272	268	258	257	252	247	240	233	228	221
B4. Other flows 3/	280	276	272	262	261	256	251	244	236	231	224
B5. One-time 30 percent nominal depreciation	280	271	251	239	234	225	216	205	195	187	178
B6. Combination of B1-B5	280	265	254	237	236	231	226	220	215	211	205
C. Tailored Tests											
C1. Combined contingent liabilities	280	306	290	277	274	268	261	254	246	241	234
C2. Natural disaster	280	315	299	287	285	279	274	267	260	256	249
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	34.9	53.3	64.7	71.1	76.4	80.3	89.9	94.1	96.9	96.9	97.8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	35	50	52	52	52	51	55	55	56	55	54
B. Bound Tests											
B1. Real GDP growth	35	54	69	78	86	93	105	111	116	117	120
B2. Primary balance	35	53	70	81	86	89	97	101	103	102	103
B3. Exports	35	53	65	71	77	81	90	94	97	98	98
B4. Other flows 3/	35	53	65	71	77	81	90	94	97	98	99
B5. One-time 30 percent nominal depreciation	35	50	62	65	72	76	85	89	91	91	92
B6. Combination of B1-B5	35	52	63	70	75	79	88	92	96	96	97
C. Tailored Tests											
C1. Combined contingent liabilities	35	53	90	95	97	99	106	109	110	109	109
C2. Natural disaster	35	54	94	100	103	105	113	116	117	116	116
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the threshold.

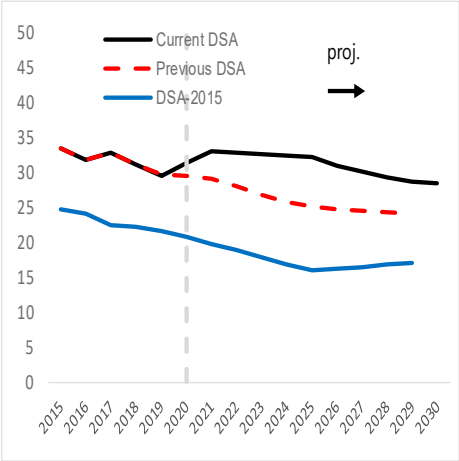
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

3/ Includes official and private transfers and FDI.

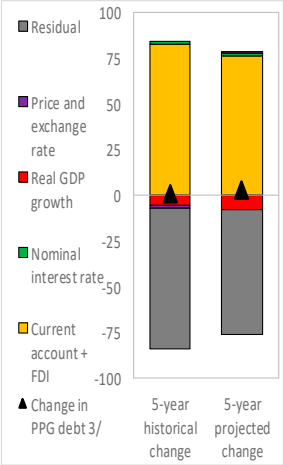
Figure 3. Malawi: Drivers of Debt Dynamics—Baseline Scenario

External debt

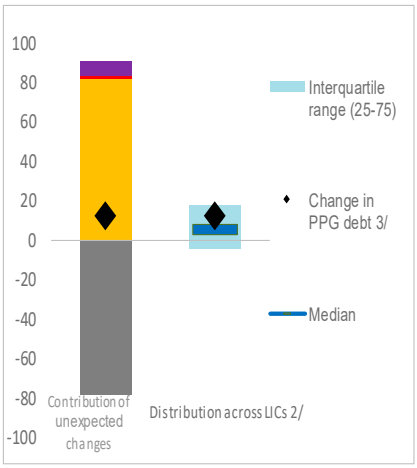
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

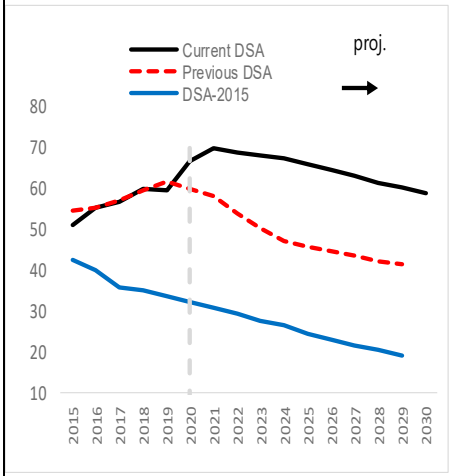


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

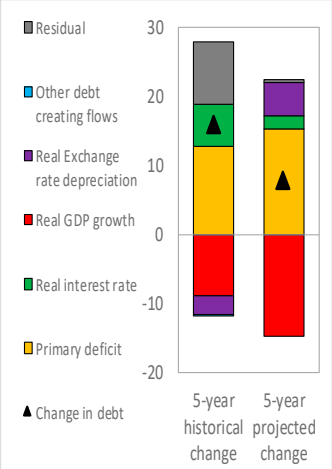


Public debt

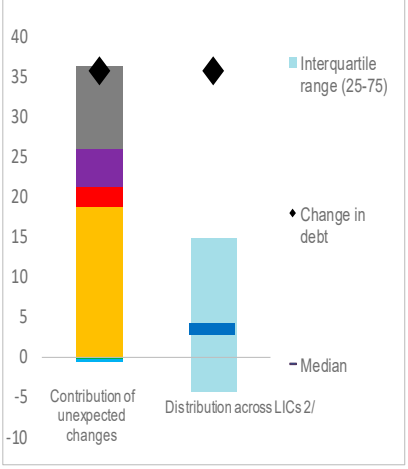
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



1/ Difference between anticipated and actual contributions on debt ratios.

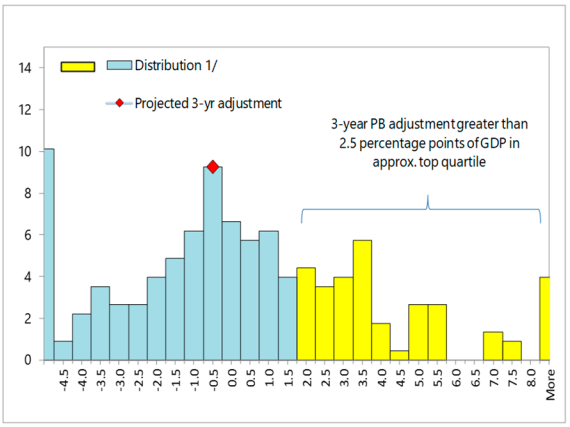
2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

Figure 4. Malawi: Realism Tools

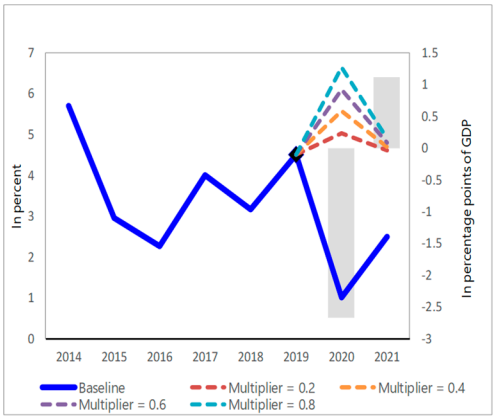
3-Year Adjustment in Primary Balance

(Percentage points of GDP)



1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

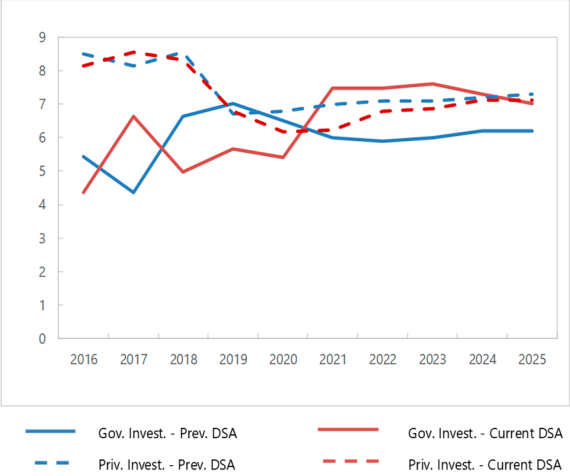
Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates

(% of GDP)



Contribution to Real GDP growth

(percent, 5-year average)

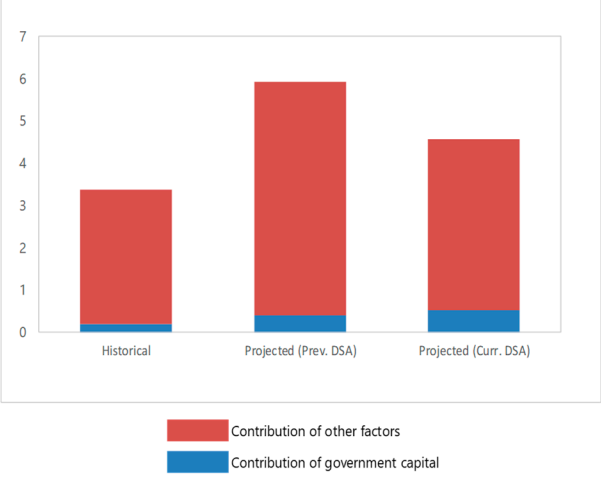
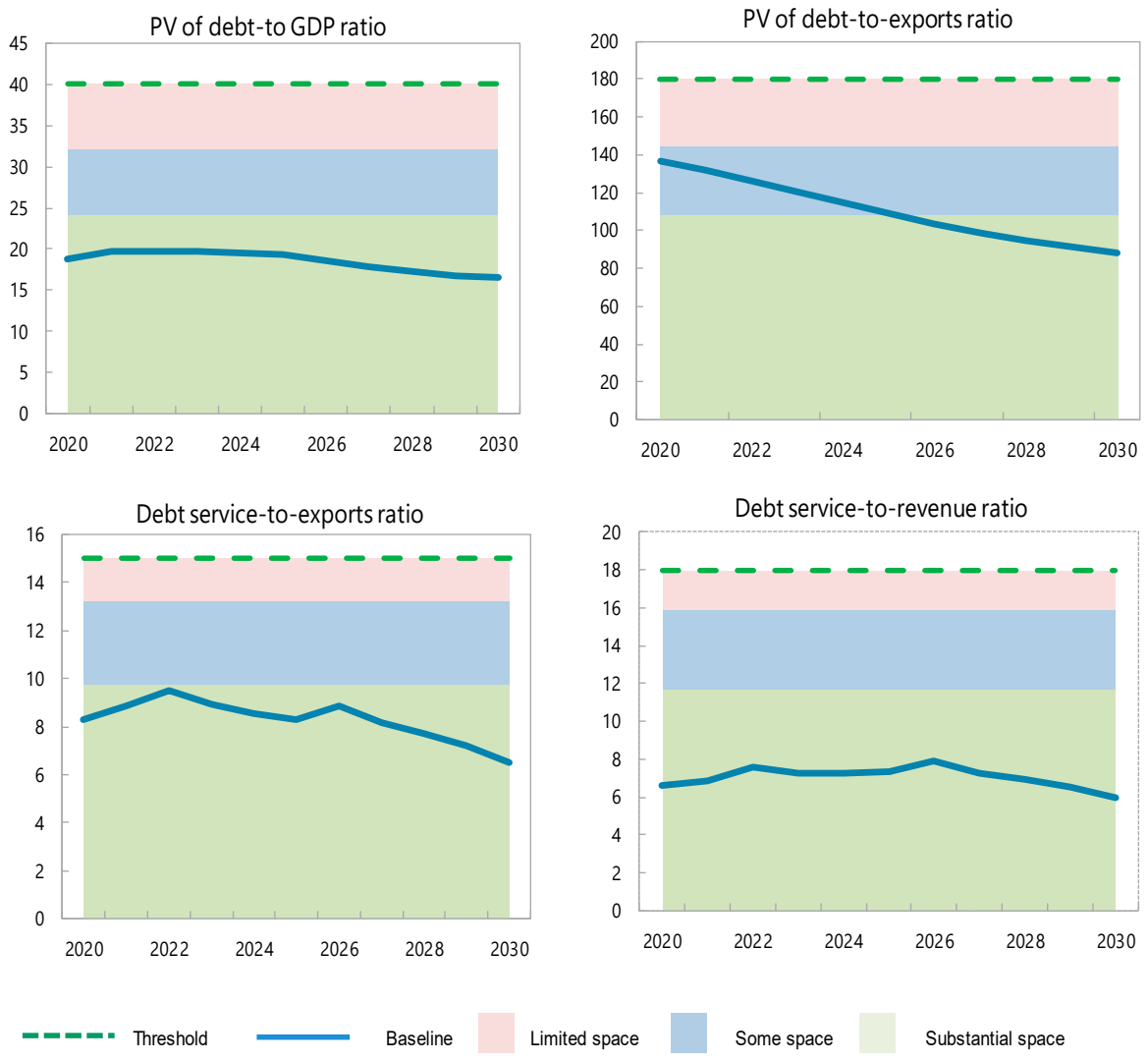


Figure 5. Malawi: Qualification of the Moderate Category, 2020–2030 1/



Sources: Country authorities; and staff estimates and projections.

1/ For the PV debt/GDP and PV debt/exports thresholds, x is 20 percent and y is 40 percent. For debt service/Exports and debt service/revenue thresholds, x is 12 percent and y is 35 percent.